That is

$$f(x)=\binom{n}{x}\left(\frac{1}{2}\right)^n, \qquad x=0,1,2,...,n.$$

The mean and variance are respectively are $\frac{n}{2}$ and $\frac{n}{4}$

Example 11.19

Find the binomial distribution for each of the following.

(i) Five fair coins are tossed once and X denotes the number of heads.

(ii) A fair die is rolled 10 times and X denotes the number of times 4 appeared.

Solution

(i) Given that five fair coins are tossed once. Since the coins are fair coins the probability of getting an head in a single coin is $p=\frac{1}{2}$ and $q=1-p=\frac{1}{2}$

Let X denote the number of heads that appear in five coins. X is a binomial random variable that takes on the values 0, 1,2,3,4 and 5, with $n=5$ and $p=\frac{1}{2}$ .That is $X\sim B\left(5,\frac{1}{2}\right)$.

Therefore the binomial distribution is

$$f(x)=\binom{n}{x}p^x(1-p)^{n-x}, \qquad x=0,1,2,...,n$$

becomes

$$f(x)=\binom{5}{x}\left(\frac{1}{2}\right)^x\left(\frac{1}{2}\right)^{n-x}, \qquad x=0,1,2,...,5\,.$$

That is

$$f(x)=\binom{5}{x}\left(\frac{1}{2}\right)^n, \qquad x=0,1,2,...,n$$

(ii) A fair die is rolled ten times and X denotes the number of times 4 appeared. X is binomial random variable that takes on the values $0, 1, 2, 3, \ldots 10$, with $n=10$ and $p=\frac{1}{6}$. That is $X\sim B\left(10,\frac{1}{6}\right)$.

Probability of getting a four in a die is $p=\frac{1}{6}$ and $q=1-p=\frac{5}{6}$.

Therefore the binomial distribution is

$$f(x)=\binom{10}{x}\left(\frac{1}{6}\right)^x\left(\frac{5}{6}\right)^{10-x}, \qquad x=0,1,2,...,10\,.$$

Example 11.20

A multiple choice examination has ten questions, each question has four distractors with exactly one correct answer. Suppose a student answers by guessing and if X denotes the number of correct answers, find (i) binomial distribution (ii) probability that the student will get seven correct answers (iii) the probability of getting at least one correct answer.

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